

ASSIGNMENT

3

Eigenvalues and
Diagonalisation

COURSE

MATH 2210 · Linear Algebra

TERM

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REMARK

Conventions. Throughout, matrices are real unless stated otherwise, and I denotes the identity matrix of the appropriate size. Show all working.

SECTION 1

Warm-up

QUESTION 1
10 MARKS

A first eigenvalue computation

Let

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

Find the eigenvalues of A .(a)
4 MARKS

Find a basis of each eigenspace.

(b)
3 MARKSIs A diagonalisable? Justify your answer.(c)
3 MARKS

(i) 1 MARK State the criterion you are using.

(ii) 2 MARKS Apply it to A .

SOLUTION

(a)

The characteristic polynomial is

$$\det(A - \lambda I) = (2 - \lambda)^2 - 1 = (\lambda - 1)(\lambda - 3),$$

so the eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = 3$.

(b) For $\lambda_1 = 1$ we solve $(A - I)v = 0$, which gives $v_1 = (1, -1)^\top$. For $\lambda_2 = 3$ we solve $(A - 3I)v = 0$, which gives $v_2 = (1, 1)^\top$.

(c) Yes.

(i) A matrix is diagonalisable if and only if $\mathbb{R}^n$ has a basis of eigenvectors.

(ii) The vectors v_1, v_2 are linearly independent, so they form a basis of $\mathbb{R}^2$ and

$$A = PDP^{-1}, \quad P = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix}.$$

■

SECTION 2

Proofs

QUESTION 2
5 MARKS

Recall the following definition.

DEFINITION 2.1

Eigenvalue. A scalar λ is an *eigenvalue* of A if $Av = \lambda v$ for some non-zero vector v .

Show that if λ is an eigenvalue of an invertible matrix A , then $\lambda \neq 0$ and λ^{-1} is an eigenvalue of A^{-1} .

HINT

Apply A^{-1} to both sides of $Av = \lambda v$.

SOLUTION

Let $Av = \lambda v$ with $v \neq 0$. If $\lambda = 0$ then $Av = 0$, which contradicts the invertibility of A . Hence $\lambda \neq 0$, and applying A^{-1} gives $v = \lambda A^{-1}v$, that is $A^{-1}v = \lambda^{-1}v$.

REMARK 2.2

The converse also holds: apply the same argument to A^{-1} .

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QUESTION 3
3+3 MARKS

Trace and determinant

Let A be a 2×2 matrix with eigenvalues λ_1, λ_2 .

(a) Prove that $\text{tr } A = \lambda_1 + \lambda_2$.

(b) Prove that $\det A = \lambda_1 \lambda_2$.

SOLUTION

Via the characteristic polynomial. Write $\chi_A(\lambda) = \lambda^2 - (\text{tr } A)\lambda + \det A$. Since λ_1, λ_2 are its roots, $\chi_A(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2)$, and comparing coefficients yields both claims.¹ ■

BONUS
2 MARKS

Does every real 2×2 matrix have a real eigenvalue? Give a proof or a counterexample.

SOLUTION

No. The rotation $R = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ has characteristic polynomial $\lambda^2 + 1$, which has no real roots. ■